

SYSTEMS ADMINISTRATION AND MAINTENANCE

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Section: BSIT – NW3B

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Activity #1 Week #7: Refresh your Knowledge

Read and review the lesson materials provided from Week #4-Week #6, from these, create a document and write your full understanding and the things you have learned in System Administration and Maintenance based on the topics and discussion.

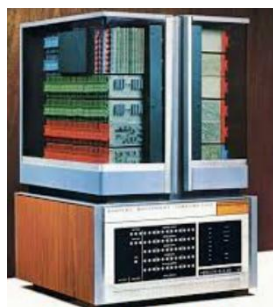
Course Packet 01

Classifications of Computer System

Computers are technological devices that process, store, and manipulate data. They are used for their capabilities in making tasks a lessening burden for the users. They are of different types, nowadays we have small computers and minicomputers with up to large mainframes, and supercomputers that are used for big production. Microcomputers are computers designed for personal use enabling communication, entertainment, and mini tasks to be done easily. Minicomputers are designed to be more affordable than mainframe computers. Mainframe Computer are large computers built for complex tasks that handles large amounts of data. Supercomputers are the fastest and not very affordable for an individual because they are utilized for scientific research and special operations.



Microcomputer



Minicomputer



Mainframe Computer



Supercomputer

Components of Computer System

Computer systems are made up of three components: hardware, software, and peopleware. Hardware is the tangible parts of the computer like keyboard, monitor, mouse, and other can be input and output data. Software comprises of application programs and software that are built with instructions that communicates with the hardware on what task to perform including operating systems and processes. Peopleware is the human user element of the computer system. It is the one that utilizes the availability of the computer. They can develop, upgrade, maintain, and input data into the computer.



Software



Hardware

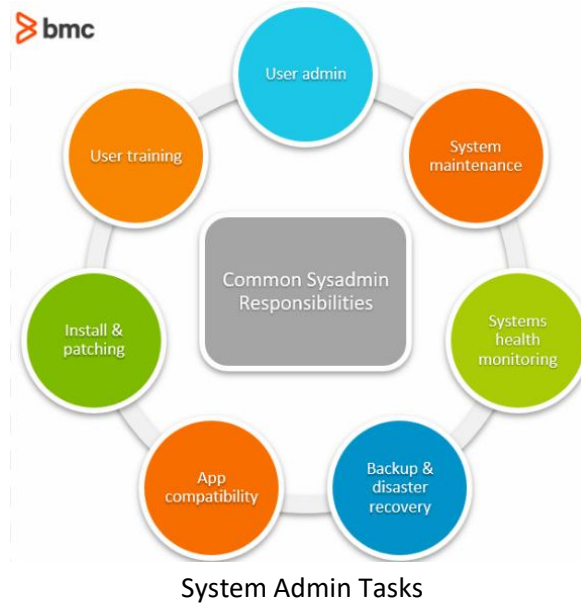


Peopleware

Course Packet 02

System Administration Overview

SysAdmins are the one that operates behind the scenes. They are the backbone of I.T. operations, ensuring security, maintains the data information, modifies functions, and configures problems that occurs. They are responsible for what may happen to the system. When in small companies they handle most of all I.T. tasks but when it comes to larger companies, they are focused on different fields to observe and maintain the performance of the system. They possess intelligent technical skills and problem-solving capabilities that adapt to developing technologies. This position requires the person to continue learning and not to be distracted by any disturbances. Take past events as a lecture for their future challenges.



Course Packet 03

Operating System Concepts, Terminology, and History

Operating Systems is the foundation of the computer system. It is the one installed first for the computer to be functional and perform operations such as read/write instructions. O.S. enables users to interact with the machine. System Administrators must be able to understand how O.S. works. It involves deep analyzation of the operating system to configure what is the standard for the users. In the history of operating systems, we have UNIX, Windows, and MacOS that offers interactive design and control administration. Understanding the past versions will help the administrators to counter the challenges and learn how to troubleshoot problems effectively.

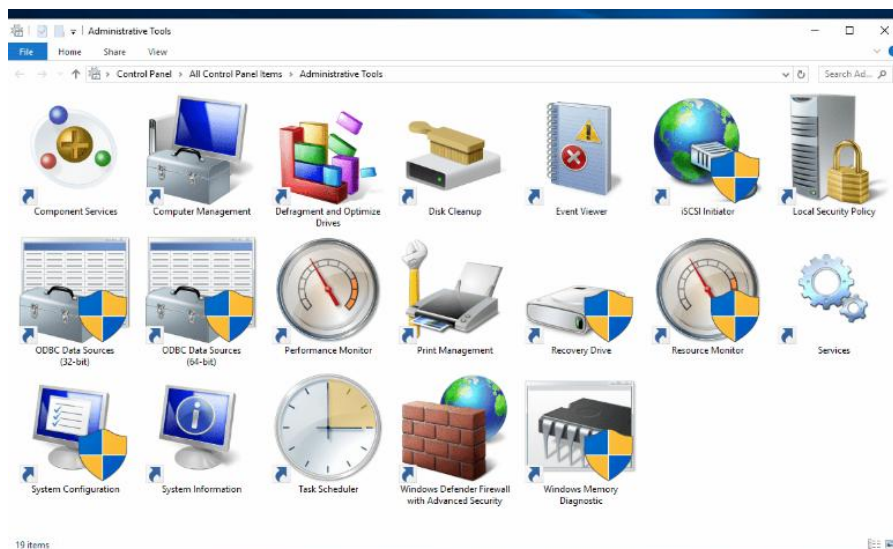


Examples of Operating System

Course Packet 04

Fundamental System Administration Tools

System Administration has major responsibilities to maintain entering and outgoing data from the system, configuring problems that trigger in some processes, and monitoring system performance. They need to be always updated on the operating trends to give the best control maintenance in the overall system. It is required that a system admin must know on how to use and maximize the GUI Tools and Power Tools. These tools help the navigation of the system to be more efficient and configure overall system structure.



Administrative Tools

Course Packet 05

IT Systems Troubleshooting and Maintenance

IT System requires a complex structure for it to be stacked organized and resolve issues in lessen difficulty. Troubleshooting involves identifying what caused it to happen and mitigating actions to prevent future issues. It includes preventive measures such as: maintenance cleaning, software updates, antivirus scan, and changing configurations of system. Maintaining a well-scheduled troubleshooting and maintenance are vital for effective system administration. Sharing ideas and continuous exchange of knowledge within the company enhances the system environment in efficiency and reliability.

Benefits Of Prioritising IT Maintenance



Early Detection
Of Problems



Improved Time
Management



Preventing Cyber
Security Attacks



Enhancing System
Performance



Maximising Software
Efficiency



Preventing
Data Loss



Documenting &
Reporting

Importance of Troubleshooting and Maintenance